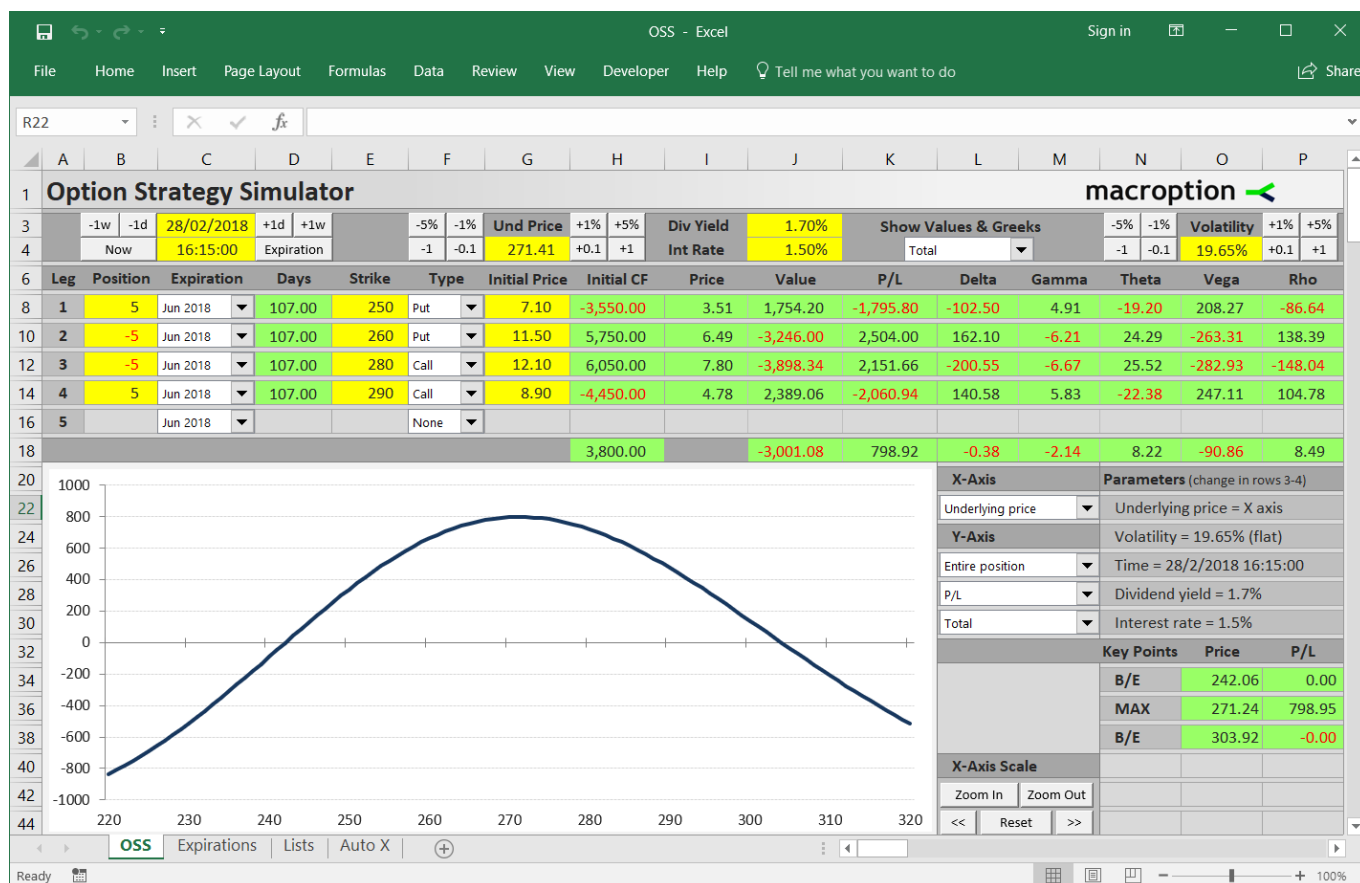


Option Strategy Simulator Guide



2 March 2018

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1. Quick Start

Hello and thank you for downloading the Option Strategy Simulator (OSS). This chapter provides essential information to get you started. The other chapters cover individual features and concepts in more detail.

Get the Right File for Your Version of Excel

There are two versions of the simulator. It is important to get the right version for your version of Excel; otherwise it won't work.

The default version is **OSS.xlsm** and it is good **for Excel 2010 and later only**.

If you have Excel 2007 or older, you must use the other file, **OSS_for_Excel_97-2007.xls**.

There are two reasons why the default version doesn't work in older versions of Excel – the NORM.DIST function (which is used for calculation of option prices and Greeks) and some conditional formatting features.

If you have just started the simulator and it is showing errors (particularly the #NAME! error), it is most likely because you are trying to run the default version (OSS.xlsm) in an older version of Excel. Please use the other file if you have Excel 2007 or older.

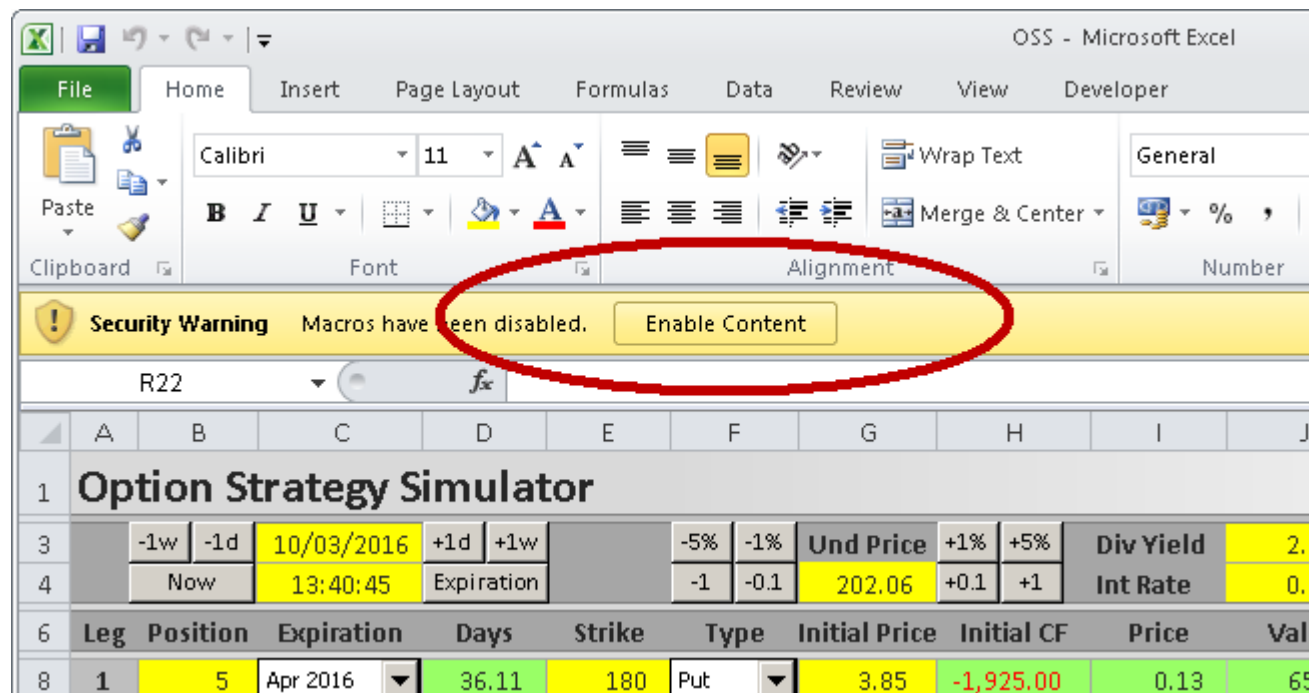
Note: Excel 2007 can run a xlsm file, but it does not support the NORM.DIST function. Therefore, if you have Excel 2007, you still need to use the xls version.

Enable Calculations and Macros

The simulator (either version) requires macros to run. **Make sure you have macros enabled** in Excel; otherwise the simulator won't work.

To protect your computer, Excel often keeps calculations and macros disabled when opening an unfamiliar file. In such case it usually displays a notice at the top, in place or below your usual Excel main menu. The notice usually says something like “Protected View – This file originated from an Internet location and might be unsafe” or “Security

Warning – Macros have been disabled”. Typically there is a button at the end of this notice, labelled “Enable Editing”, “Enable Content” or similar. Click on this button.



The screenshot above is from Excel 2010. The message and the button might look different in your computer, depending on your Excel version and settings.

Sheets

There are four sheets in the file. Users work with the first two sheets only:

- **“OSS”** – This is the main sheet where almost everything important happens. Here you can set and change parameters, run simulations and see option prices, profit or loss, Greeks and chart. Most of the time you will work with this sheet only. You can find an overview of this sheet’s layout and features in [chapter 2](#) and more details in subsequent chapters.
- **“Expirations”** – In this sheet you can enter and edit option expiration dates and adjust the way they are displayed. You can find more details in [chapter 4](#).

The other two sheets – **“Lists”** and **“Auto X”** – are not supposed to be changed by users. If you do, the simulator may not work properly. The former stores values for the combos. The latter calculates scale of the chart’s X axis.

Key Rule: Only Overwrite Yellow Cells

You will find cells with three different background colours in the main user areas:

- **Yellow cells** are for user input.
- **Green cells** are for output (results).
- **Grey cells** are labels or additional information.

For example:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Option Strategy Simulator											
3	-1w	-1d	10/03/2016	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	2.05%
4	Now		13:40:45	Expiration		-1	-0.1	202.06	+0.1	+1	Int Rate	0.50%
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta
8	1	5	Apr 2016	36.11	180	Put	3.85	-1,925.00	0.13	65.45	-1,859.55	-13.72
10	2	-5	Apr 2016	36.11	190	Put	5.95	2,975.00	1.00	-498.32	2,476.68	76.34
12	3	-5	Apr 2016	36.11	210	Call	4.70	2,350.00	1.89	-943.63	1,406.37	-131.37
14	4	5	Apr 2016	36.11	220	Call	1.05	-525.00	0.43	216.33	-308.67	40.03
16	5		Apr 2016			None						
18								2,875.00		-1,160.17	1,714.83	-28.72

Only change yellow cells. You can of course also change the selection in combos.

Do not overwrite the contents of any other cells (green, grey and other colours).

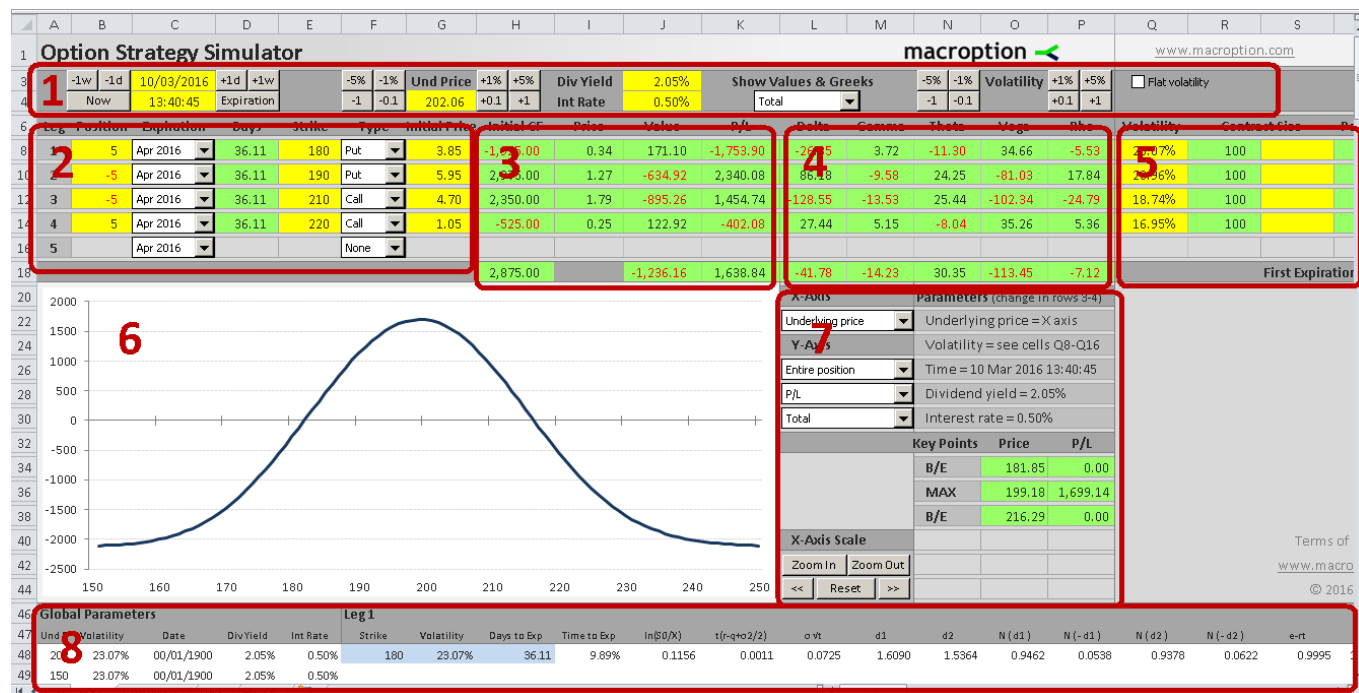
This rule applies throughout the whole simulator (and throughout all other Macroption Excel files).

Besides the three main colours you can find a few cells with other colours in the back end calculation areas. Different colours demark cells with different formula patterns or cells of special importance. Most users don't need to worry about and should not change these cells.

That said, feel free to explore all formulas, change cells of any colour, and adapt the file to your own needs – at your own risk. It is always best to make a backup copy of the original file before making any changes.

2. Main Sheet Overview

This chapter provides basic orientation in the main sheet, which is named “OSS”. The screenshot below marks eight areas:



Area 1 (rows 3-4) = parameters – Here you can enter and change the main parameters, i.e. valuation date, underlying price, dividend yield, interest rate and volatility. It should be quite obvious, but you can find detailed instructions in [chapter 5](#).

Area 2 (cells B8-G16) = position – Here you can specify up to five holdings (legs) as part of your strategy. Each of the five legs can be set as Call, Put, Stock (underlying) or None. For each you need to set position size, expiration, strike and initial price, if applicable. You can find more details in [chapter 3](#).

Area 3 (cells H8-K18) = valuation – Here you can view the calculated value and profit or loss for each individual leg and for the entire position, either per share or total. More in [chapter 6](#).

Area 4 (cells L8-P18) = Greeks – Here you can see the Greeks (delta, gamma, theta, vega and rho), again for individual legs and for the entire position, either per share or total. More in [chapter 6](#).

Area 5 (cells Q8-V18) = leg custom settings – Here you can, optionally, set specific settings for individual legs, namely volatility (if you choose to not use the same volatility input for all legs) and contract multiplier. More in [chapter 3](#).

Area 6 (cells A20-K44) = chart

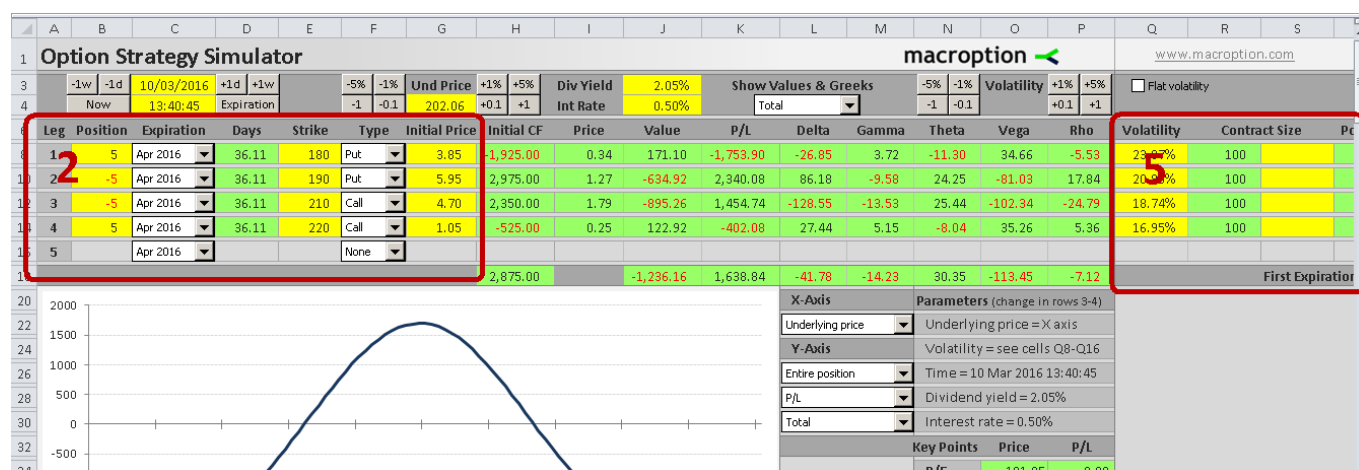
Area 7 (cells L20-P44) = chart settings and values – On the left in columns L-M you can choose what you want to see in the chart (top) and adjust X axis scale (bottom). On the right in columns N-P you can see a recap of the global parameters used in the chart (top) and below that an overview of key points (break-evens, minimums and maximums). More in [chapter 7](#).

Area 8 (row 46 and below) = calculations – The various calculations powering all the above listed features are done in rows 48 and below. You should not change anything in this area, unless you know what you are doing and want to change how the simulator works. If you want to explore the formulas and concepts behind these calculations, you can find more information in [chapters 8, 9 and 10](#).

When working with the simulator, always keep the core rule in mind – you can change yellow cells and combos, but you should not overwrite the content of other cells.

3. Entering Position

This chapter provides detailed explanation of all the inputs for individual legs, such as position size, expiration and strike. It covers the areas which were marked as 2 and 5 in the [previous chapter](#).



Selecting Leg Type

You can enter up to five legs in rows 8, 10, 12, 14, 16.

Start with selecting **Type** in the combo in **column F**. The four available types are:

- None – Select this one if you won't use this particular leg.
- Stock – Select this one for long or short position in the underlying security. It is named Stock, but also works for other assets, such as ETFs, futures, currencies or bonds.
- Call – Select this one for long or short call option position.
- Put – Select this one for long or short put option position.

Other cells in rows 8-16 are formatted based on the Type selection. Yellow cells are where you should enter other position settings (like number of contracts or initial price). Green cells are where you can see results (do not overwrite these). Grey cells are those which don't apply for the particular position type (for instance, if you select Stock, the cells for strike or days to expiration will be grey).

The order of legs does not matter for the calculations and results. You may want to order them by strikes or expirations for convenience, but you can also enter them in completely random order.

Entering a Call or Put Option Position

If you select Call or Put as Type, there will be four other inputs you need to enter for the particular leg.

- Expiration date
- Strike price
- Position size (number of contracts)
- Initial price

	A	B	C	D	E	F	G	H	I				
1	Option Strategy Simulator												
3		-1w	-1d	10/03/2016	+1d	+1w		-5%	-1%	Und Price	+1%	+5%	Div Yield
4		Now		13:40:45	Expiration			-1	-0.1	202.06	+0.1	+1	Int Rate
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price				
8	1	5	Apr 2016	36.11	180	Put	3.85	-1,925.00	0.34				
10	2	-5	Apr 2016	36.11	190	Put	5.95	2,975.00	1.27				
12	3	-5	Apr 2016	36.11	210	Call	4.70	2,350.00	1.79				
14	4	5	Apr 2016	36.11	220	Call	1.05	-525.00	0.25				
16	5		Apr 2016			None							
18									2,875.00				

Select **expiration** in the combo in **column C**. The expirations preset as default are US stock option monthly expirations. If you need other expirations or want to change how the expiration dates are displayed (e.g. add day of month), you can edit everything in the sheet "Expirations". You can find detailed instructions in [chapter 4](#).

Based on your selection in the combo in column C and the valuation date and time set in cells C3-C4 (more on that later), you will see the number of days left to expiration in column D (green cell – do not overwrite it). For example, for the standard monthly US stock options, which cease trading at close (16:15) on Friday, the valuation date and time set to 14:30 the Thursday before that will show 1.07 days left to expiration.

Next you need to enter the **strike price**, typing the value in **column E**. Make sure it is entered in the same units (with same contract multiplier) as the underlying price (cell G4) – this is usually very obvious, but may be less clear with some kinds of options, such as currency options or futures options.

Enter your **position size** (number of contracts) by typing the value in **column B**. Enter it **with negative sign for a short position**. For example, 5 in column B with Call selected in column F means long 5 call option contracts; -5 means short 5 call option contracts.

By default, the **contract multiplier** is preset at 100 shares (one option contract represents the right to buy or sell 100 shares of the underlying), which is for standard US stock and ETF options and many other option contracts around the world. To work with options with different contract size, you can change the multiplier either globally or just for the individual leg (if you have different kinds of options with different contract multipliers in different legs).

To change the default contract multiplier **globally**, edit it in **cell Y8**.

To change the contract multiplier **just for one leg**, override the default by entering the new multiplier in **column S** in the particular row.

	P	Q	R	S	T	U	V	W	X	Y							
1		www.macroption.com					support@macroption.com										
3	+1%	+5%	☐ Flat volatility														
4	+0.1	+1															
6	Rho	Volatility	Contract Size		Position	Expiration		Name		<table><tr><th colspan="2">Default Contract Size</th></tr><tr><td>Options</td><td>100</td></tr><tr><td>Stock</td><td>1</td></tr></table>		Default Contract Size		Options	100	Stock	1
Default Contract Size																	
Options	100																
Stock	1																
8	-55.19	23.07%	1,000	1,000	5,000	15/04/2016 16:15:00		+5 15Apr 180P									
10	17.81	20.96%	100		-500	15/04/2016 16:15:00		-5 15Apr 190P									
12	-24.76	18.74%	100		-500	15/04/2016 16:15:00		-5 15Apr 210C									
14	5.35	16.95%	100		500	15/04/2016 16:15:00		+5 15Apr 220C									
16																	
18	-56.79	First Expiration				15/04/2016 16:15:00											

You can always see the contract multiplier which is currently being used for each leg in column R. If the cell in column S is empty, the default from cell Y8 is used.

In column T you can see the total number of shares your position represents. For example, if you have entered 5 in column B (long 5 option contracts) and 1,000 as contract multiplier in column S, it will show 5,000 (long 5,000 shares), as shown in row 8 in the screenshot.

The fourth input for a particular leg is **initial price**, entered in **column G**. This is the average price you have paid when buying or received when selling the options. It should be entered per share (or generally in the same units as strike price and underlying price) and always with positive sign (unless you have traded in and out of the position and made profits high enough for a positive total cash flow with an existing long position or losses high enough for a negative total cash flow with a short position – in that case enter it with minus sign). Typically you will also include any commissions.

If you are not using the simulator to analyse a position you are actually trading or considering as a potential trade and don't have any values for initial price, you can keep the cells in column G empty. Everything will work the same, except the profit and loss (P/L) value, which will be the same as position value (as it will assume initial cash flow is zero).

Entering a Long or Short Position in the Underlying

For some strategies (such as covered calls or protective puts) you may also want to include a long or short position in the underlying security. Select "Stock" in the combo in column F.

The expiration date combo in column C is still visible, but it has no effect and you don't need to worry about it. Strike price does not apply either.

Enter **position size** in **column B**, again a positive number for a long underlying position and negative for a short position. For example, if you are long 745 shares in the underlying stock, you will enter 745.

If your underlying security trades in contracts rather than individual shares (e.g. when trading futures and options on futures), you can change the **contract multiplier** in the same way as for option contracts, either globally in cell Y10 or for an individual leg in column S. In such case you need to enter number of contracts, instead of number of shares, as position size in column B, like you do for options.

You can also enter **initial price** in **column G** in the same way as for option positions – same units as the underlying price and positive sign for both long and short positions.

Examples

The following screenshots demonstrate how the leg inputs look for some of the popular option strategies.

Although some people would not consider it a “strategy”, you can also use the simulator for analysing **single option positions**. The one below is long 5 contracts of 70 strike calls, expiring in April 2016, for which you have paid \$1.36 per contract, with total initial cost of \$680. All the other legs are set to None.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1	5	Apr 2016	36.18	70	Call	1.36	-680
10	2		Apr 2016			None		
12	3		Apr 2016			None		
14	4		Apr 2016			None		
16	5		Apr 2016			None		
18								-680

Covered Call = Long position in the underlying and a corresponding short call option position. Leg 1 is the short call (10 contracts, 210 strike, sold for \$2.71 per share). Leg 2 is the long underlying position (1,000 shares, bought for average price of \$202.38 per share). The order of legs doesn’t matter, although just for convenience or out of habit most people would probably prefer to put the underlying position first.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1	-10	Apr 2016	36.18	210	Call	2.71	2,710
10	2	1,000	Apr 2016			Stock	202.38	-202,380
12	3		Apr 2016			None		
14	4		Apr 2016			None		
16	5		Apr 2016			None		
18								-199,670

Protective Put = Protecting a long position in the underlying (leg 1) with long put options (leg 2). In this particular case, the number of shares does not exactly match, as 23 contracts represent 2,300 shares, 15 less than the long stock position. Average cost of the long underlying position is \$31.46 per share; the puts were bought for \$1.13 per share.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1	2,315	May 2016			Stock	31.46	-72,830
10	2	23	May 2016	65.24	30	Put	1.13	-2,599
12	3		Apr 2016			None		
14	4		Apr 2016			None		
16	5		Apr 2016			None		
18								-75,429

Short Call Spread (also known as Bear Call Spread) = Selling a lower strike call (leg 1) and buying a higher strike call (leg 2). 5 contracts of each, May 2016 expiration, strikes 32.5/35. The short options sold for \$1.59 per share; the long bought at \$0.84.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1	-5	May 2016	51.27	32.5	Call	1.59	795.00
10	2	5	May 2016	51.27	35	Call	0.84	-420.00
12	3		Apr 2016			None		
14	4		Apr 2016			None		
16	5		Apr 2016			None		
18								375.00

Long Call Calendar Spread = Buying a call with longer expiration (June 2016, leg 1) and selling a call with the same strike (30), but shorter expiration (April 2016, leg 2). 8 contracts of each. Longs bought at \$2.63; shorts sold at \$1.57.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1	8	Jun 2016	93.00	30	Call	2.63	-2,104.00
10	2	-8	Apr 2016	30.00	30	Call	1.57	1,256.00
12	3		Apr 2016			None		
14	4		Apr 2016			None		
16	5		Apr 2016			None		
18								-848.00

Long Straddle = Long call (leg 2) and long put (leg 3) with same strikes and same expirations. Again, the order of legs does not matter in the simulator and you don't need to start from leg 1.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1		Jun 2016			None		
10	2	8	May 2016	65.00	30	Call	1.87	-1,496.00
12	3	8	May 2016	65.00	30	Put	1.95	-1,560.00
14	4		Apr 2016			None		
16	5		Apr 2016			None		
18								-3,056.00

Iron Condor = Short calls (leg 3) and puts (leg 2) with different strikes, long higher strike calls (leg 4) and lower strike puts (leg 1). Same expiration and position size for all.

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1	5	Apr 2016	22.13	190	Put	0.55	-275.00
10	2	-5	Apr 2016	22.13	200	Put	2.58	1,290.00
12	3	-5	Apr 2016	22.13	210	Call	3.28	1,640.00
14	4	5	Apr 2016	22.13	220	Call	0.93	-465.00
16	5		Apr 2016			None		
18								2,190.00

Long Ratio Put Spread = This is an example of strategies where the position sizes of individual legs are different. The number of contracts in leg 4 (long 80 puts) is double the number of contracts in leg 3 (short 85 puts). In this particular case the initial cash flow is similar for both legs and total initial cash flow is close to zero (\$45).

	A	B	C	D	E	F	G	H
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF
8	1		Apr 2016			None		
10	2		Apr 2016			None		
12	3	-3	May 2016	51.00	85	Put	4.25	1,275.00
14	4	6	May 2016	51.00	80	Put	2.05	-1,230.00
16	5		Apr 2016			None		
18								45.00

4. Updating Expiration Dates

As the [previous chapter](#) has illustrated, expiration dates for individual legs are selected in drop-down boxes in column C rather than typed in manually. This should be faster and less prone to errors in the long run, as users typically work with the same set of expiration dates over and over again.

That said, sometimes you may need to update the preset expiration dates, either as the existing ones expire and new ones become available, or as you work with a different kind of options with different expiration dates.

The values which appear in the drop-down boxes in column C in the sheet “OSS” are stored in a separate sheet, named “Expirations”. In this sheet you can change the expiration dates and also change the way the dates are displayed.

	A	B	C	D	E	F	G	H	I	J	K
1											
2			Auto Name Format	mmm yyyy		dmmm					
3											
4											
5			ID	Expiration Date+Time	Auto Name 1	Custom Name 1	Auto Name 2	Custom Name 2			
6			1	19/01/2018 16:15:00	Jan 2018		19Jan				
7			2	16/02/2018 16:15:00	Feb 2018		16Feb				
8			3	16/03/2018 16:15:00	Mar 2018		16Mar				
9			4	20/04/2018 16:15:00	Apr 2018		20Apr				
10			5	18/05/2018 16:15:00	May 2018		18May				
11			6	15/06/2018 16:15:00	Jun 2018		15Jun				
12			7	20/07/2018 16:15:00	Jul 2018		20Jul				
13			8	17/08/2018 16:15:00	Aug 2018		17Aug				
14			9	21/09/2018 16:15:00	Sep 2018		21Sep				
15			10	19/10/2018 16:15:00	Oct 2018		19Oct				
16			11	16/11/2018 16:15:00	Nov 2018		16Nov				
17			12	21/12/2018 16:15:00	Dec 2018		21Dec				
18			13	18/01/2019 16:15:00	Jan 2019		18Jan				
19			14	15/02/2019 16:15:00	Feb 2019		15Feb				
20			15	15/03/2019 16:15:00	Mar 2019		15Mar				
21			16	18/04/2019 16:15:00	Apr 2019		18Apr				
22			17	17/05/2019 16:15:00	May 2019		17May				
23			18	21/06/2019 16:15:00	Jun 2019		21Jun				
24			19	19/07/2019 16:15:00	Jul 2019		19Jul				
25			20	16/08/2019 16:15:00	Aug 2019		16Aug				
26			21	20/09/2019 16:15:00	Sep 2019		20Sep				
27			22	18/10/2019 16:15:00	Oct 2019		18Oct				
28			23	15/11/2019 16:15:00	Nov 2019		15Nov				
29			24	20/12/2019 16:15:00	Dec 2019		20Dec				

Auto Name Formats	Example
1 mmm yy	Mar 18
2 mmm yyyy	Mar 2018
3 mm/dd/yy	03/02/18
4 dd/mm/yy	02/03/18
5 m/d/yy	3/2/18
6 d/m/yy	2/3/18
7 m/d	3/2
8 d/m	2/3
9 dmmm	2Mar
10	

The screenshot above shows how the sheet looks. There are two tables. The larger one on the left (cells B2-G28) stores the actual expiration dates. The smaller table on the right (cells I4-K14) stores available date formats.

Adding a New Expiration

To **add a new expiration**, simply enter the new expiration date and time in one of the available yellow cells in **column C (C5-C28)** or replace an existing entry. There is space for 24 different expirations (for instance, two years of monthly expirations).

Because the simulator works with intraday option pricing, it is very important to **also enter the correct expiration time**.

The timezone should be your local time (or more precisely, the default timezone used by your computer's operating system), which may be different from the timezone of the options exchange. For example, if you are in the UK and trade US stock options, the expiration time should be 21:15, which is the local time in the UK when US options stop trading (16:15 New York time, 15:15 Chicago time). Also note the time difference between some timezones (e.g. US vs. UK/Europe) can be different in some parts of the year due to different Daylight Saving Time start and end dates.

Furthermore, the date and time to enter as "expiration date" in column C is always the date and time **when the options stop trading, when a settlement price is determined** or when an option's holder is effectively last able to exercise (or more precisely, when no further moves in the underlying price can potentially affect the option's final settlement value). In some cases this may not be exactly the same date as the "expiration date" presented by the particular options exchange for the contract.

For example, until recently the expiration of US standard monthly options was presented as "the Saturday after the third Friday of the month", but effectively for the trader it was the Friday market close when it all stopped and that was also the moment to use as expiration in option pricing models. Electronic trading finally allowed CBOE to change this in 2015 and now standard monthly options officially expire on Fridays, the same day as when they stop trading, which should also be entered in column C. That said, some other exchanges or some OTC options may still have officially stated expiration dates different from the effective expiration dates which should be used here.

Expiration Date Formats

The **other columns in the large table (D-G)** decide how the expiration dates are displayed in the simulator. There are two different active formats at a time.

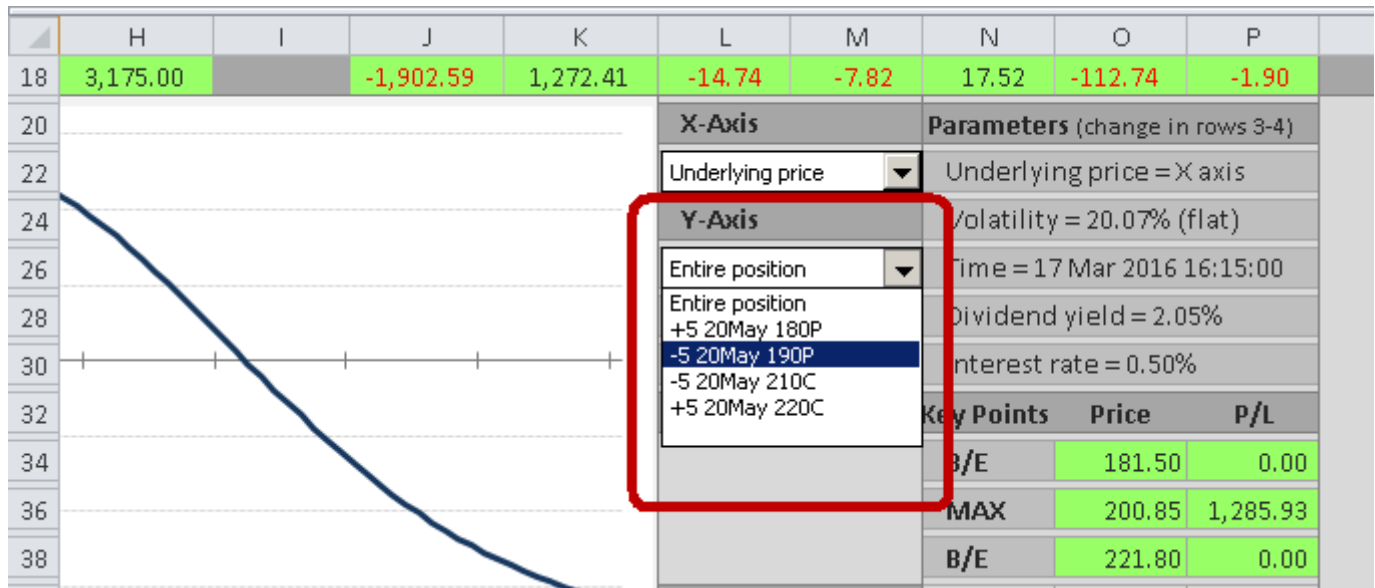
Name 1 is how the expirations are seen in the combo boxes in the leg inputs in column C in the sheet “OSS” (example below).

	A	B	C	D	E	F	G	H	I	J	K
1	Option Strategy Simulator										
3	-1w	-1d	17/03/2016	+1d	+1w		-5%	-1%	Und Price	+1%	+5%
4	Now		16:15:00	Expiration			-1	-0.1	202.50	+0.1	+1
6	Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L
8	1		Apr 2016	29.00	190	Put	0.55	-275.00	0.87	433.57	158.5
10	2		May 2016	29.00	200	Put	2.58	1,290.00	3.70	-1,852.44	-562.4
12	3		Jun 2016	29.00	210	Call	3.28	1,640.00	1.94	-969.93	670.0
14	4		Jul 2016	29.00	220	Call	0.93	-465.00	0.44	218.03	-246.9
16	5		Aug 2016			None					
18			Sep 2016								
20			Oct 2016								
22			Nov 2016								
24			Dec 2016								
26			Jan 2017								

By default the format of Name 1 is “mmm yyyy”, i.e. the first three letters of month and full year, for example “Jan 2016” for options expiring 15 January 2016. You can change the format in the drop-down box in cell D2 (in the sheet “Expirations”) or enter your own custom name for the particular expiration in column E (E5-E28). If the cell in column E is empty, the format selected in cell D2 is used for that particular expiration.

If you trade weekly options or have multiple expirations within the same month, you may want to use a format which also includes day of month.

Name 2 is how the expirations are displayed when referring to an entire leg, i.e. including position size, strike and option type. For instance, you can see this in the drop-down box for selecting the chart’s Y axis in cell M26 in the sheet “OSS” (example on the next page).



It is best to keep Name 2 shorter. The default is “dmmm”, e.g. “15Jan” for 15 January 2016. Like for Name 1, you can change the format for Name 2 in the combo box in cell F2 in the sheet “Expirations” or enter custom names for individual expirations in column G (G5-G28).

The formats available in the drop-down boxes in cells D2 and F2 are those entered in **column J (J5-J14)**, which you can adjust to your needs. In column K (K5-K14) you can see how today’s date looks with each format.

	H	I	J	K	L
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

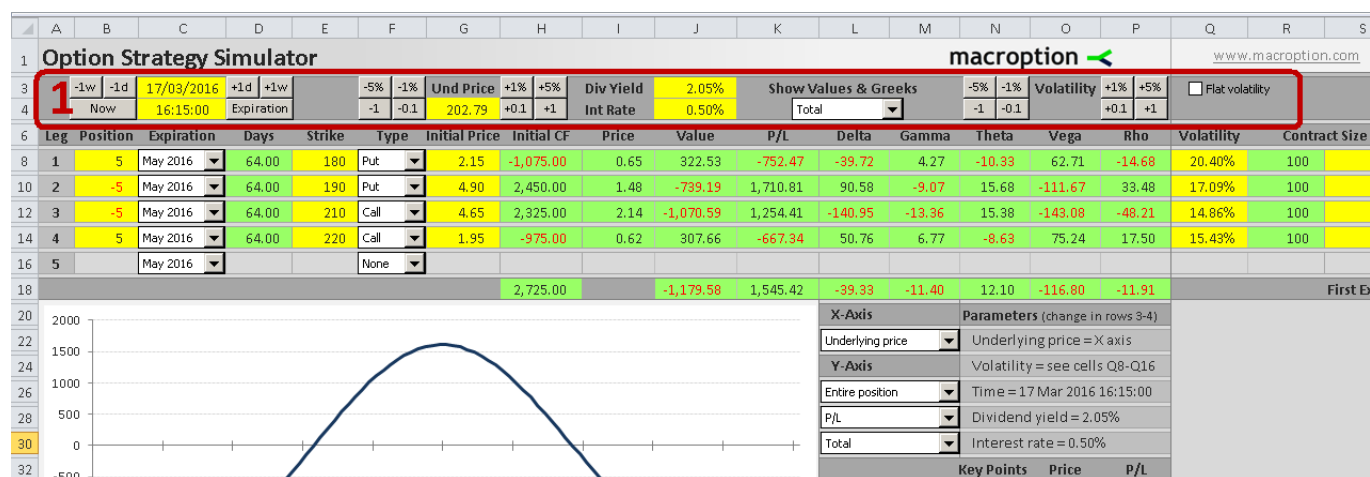
	Auto Name Formats	Example
1	mmm yy	Mar 16
2	mmm yyyy	Mar 2016
3	mm/dd/yy	03/18/16
4	dd/mm/yy	18/03/16
5	m/d/yy	3/18/16
6	d/m/yy	18/3/16
7	m/d	3/18
8	d/m	18/3
9	dmmm	18Mar
10		

5. Parameters

This chapter explains how to enter and change the option pricing parameters, which you can find in rows 3-4 of the main sheet. Going from left to right they are:

- Valuation date and time
- Underlying price
- Dividend yield
- Interest rate
- Volatility

It is the area which was marked as 1 in [chapter 2](#).



Valuation Date and Time

This is the date and time for which you want to see option prices, profit or loss, Greeks and the other statistics. It will often be the current date and time, but you can set it to any time in the future or in the past. The number of days remaining to expiration, which you can see in column D for legs which are options, is calculated as the positive difference between the particular option's expiration and valuation date and time.

Valuation date is entered in the yellow **cell C3**. Time is entered in **cell C4**.

There are two ways to change these – either by directly typing a new date and time in the cells or by clicking one of the surrounding buttons.

The buttons in the upper row (row 3) are “-1w” (one week back), “-1d” (one day back), “17/03/2016” (one day forward) and “+1d” (one day forward) and “+1w” (one week forward).

The buttons in the lower row (row 4) are “Now” (current date and time) and “Expiration” (option expiration date and time). If your position includes options with different expiration dates, clicking the “Expiration” button will set valuation date to the first (oldest) expiration date among your legs.

	A	B	C	D	E
1	Option Strategy Simulator				
3		-1w	-1d	17/03/2016	+1d +1w
4		Now	10:06:21	Expiration	
6	Leg	Position	Expiration	Days	Stri
8	1	5	May 2016	64.26	1
10	2	-5	May 2016	64.26	1
12	3	-5	May 2016	64.26	2
14	4	5	May 2016	64.26	2
16	5		May 2016		
18					

Underlying Price

Underlying price is the price at which the underlying security is trading on the market at the moment you are doing the valuation. It should have the same units as strike prices of option legs – usually (for stock or ETF options) per share.

Underlying price is entered in the yellow cell **G4**. There are again two ways to change it – by typing a value in the cell or by clicking buttons.

	E	F	G	H	
1	Underlying Price				
3		-5%	-1%	Und Price	+1% +5% Div Y
4		-1	-0.1	202.79	+0.1 +1 Int R
6	Strike	Type	Initial Price	Initial CF	Pi
8	180	Put	2.15	-1,075.00	
10	190	Put	4.90	2,450.00	
12	210	Call	4.65	2,325.00	
14	220	Call	1.95	-975.00	
16		None			
18				2,725.00	

The meaning of buttons should be obvious. The upper ones change the underlying price by a certain percentage, while the lower ones change it by a dollar amount.

Dividend Yield

Dividend yield is entered by typing a value in the yellow cell **J3**. The units are percent per annum, continuously compounded. This is one of the ways how the option pricing model might deviate from the reality of your trading, as most underlying securities will pay discrete dividends (a certain amount on a certain date, one or several times per year).

Also note that the use of this parameter is not limited to stocks and dividends. You can also use it for other types of underlyings which pay interest (such as currencies or bonds) or convenience yield (like some commodities).

There are no buttons available, but you probably won't be changing the dividend yield as often as the parameters above.

	I	J	K	L	M
1	n				
3	Div Yield	2.05%	Show Values & Greeks		
4	Int Rate	0.50%	Total		
6	Price	Value	P/L	Delta	Gamma
8	0.65	325.18	-749.82	-39.92	4.27
10	1.49	-743.20	1,706.80	90.82	-9.06
12	2.15	-1,074.52	1,250.48	-141.13	-13.34
14	0.62	309.87	-665.13	50.98	6.77
16					
18		-1,182.67	1,542.33	-39.25	-11.36

Interest Rate

Risk-free interest rate is entered by typing a value in the yellow cell **J4**, right below dividend yield. There are no buttons. The units are again percent per annum, continuously compounded.

The interest rate's tenor (time to maturity) should approximately match the time to expiration of the options you are pricing. This parameter tends to have far lower effect than some of the others, especially in today's low interest rate environment, but some particular strategies can be quite sensitive to changes in interest rates, usually depending on the selection of strikes.

If you are working with forex options, where you have two different interest rates for the two currencies, enter the "home" interest rate here (the one in which the options are denominated, typically the USD rate for US traded forex options). The "foreign" interest rate can be entered as dividend yield, because mathematically it is the same thing (compounding frequency aside).

Continuing to the right, you will find a drop-down box next to the dividend yield and interest rate settings, labelled “Show Values & Greeks”. This is not an option pricing parameter, but it allows you to choose how to display the results (individual leg and total position values, profit or loss and Greeks) – either per share or for your actual position size. This will be covered in more detail in [chapter 6](#).

Volatility

Implied volatility of different options on the same underlying (with different strikes or different expirations) is often different – you have probably heard the terms “volatility skew” or “volatility surface”. The simulator allows you to **choose whether you want to use different volatility input for each leg, or use the same “flat” volatility for the entire position**. The latter is obviously much easier to work with. Whether you can afford this simplification depends on the degree of precision you need and the underlying you are working with (how far from flat its volatility pattern is). The selection is made by ticking the box in **cell Q3**.

If the “Flat volatility” box is checked, the same volatility will be used for all legs. In such case, volatility is entered in the yellow **cell O4**, either by overwriting the value directly or by clicking one of the surrounding buttons.

Volatility should be entered as percentage and always annualized. Like most other popular option resources, the simulator works with volatility as standard deviation (not variance).

	N	O	P	Q	R
1	macroption 			www.macroption.com	
3	-5%	-1%	Volatility	+1%	+5%
4	-1	-0.1	20.41%	+0.1	+1
6	Theta	Vega	Rho	Contra	
8	-10.35	62.76	-14.70		100
10	20.87	-125.10	40.87		100
12	23.35	-155.87	-57.79		100
14	-16.50	108.21	29.27		100
16					
18	17.37	-110.00	-2.35		

The buttons in the upper row change the volatility input by a certain percentage. Those in the lower row change it by a certain number of percentage points (volatility points). If you are not sure you understand the difference, it is best to play with the buttons for a while and see what they do.

If the “Flat volatility” box is unchecked, cell O4 is grey and won’t work. Instead, you will enter the individual volatility inputs for individual legs in **column Q**. The buttons will still work – clicking one will change all the individual leg volatilities by the same amount or percentage.

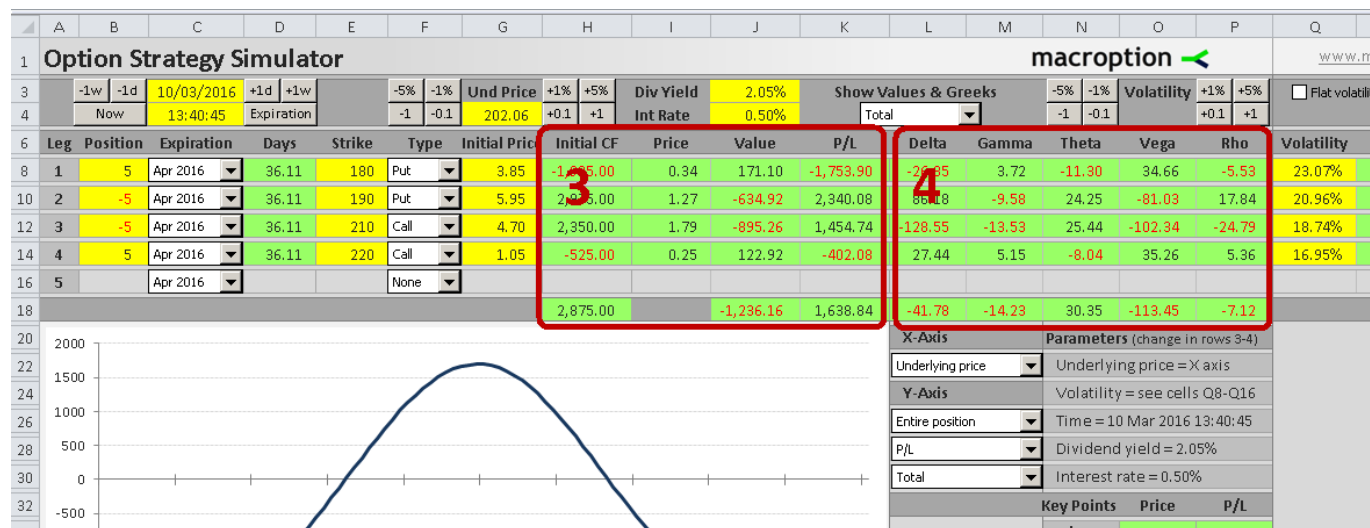
	M	N	O	P	Q	R	S	T
1	macroption 				www.macroption.com			
3	Weeks	-5%	-1%	Volatility	+1%	+5%	☐ Flat volatility	
4		-1	-0.1		+0.1	+1		
6	Gamma	Theta	Vega	Rho	Volatility	Contract Size	Position	
8	4.27	-10.33	62.71	-14.68	20.40%	100		500
10	-9.07	15.68	-111.67	33.48	17.09%	100		-500
12	-13.36	15.38	-143.08	-48.21	14.86%	100		-500
14	6.77	-8.63	75.24	17.50	15.43%	100		500
16								
18	-11.40	12.10	-116.80	-11.91		First Expiration		20

Volatility is the most challenging and most important parameter when it comes to option pricing and trading. Even small changes in volatility can have huge effects on option prices – and profits or losses. At the same time, unlike time to expiration or underlying price, volatility is unknown and must be estimated.

There are various methods to estimate volatility, considering factors such as historical (realized) volatility, implied volatility of similar options, fundamentals of the underlying asset or seasonality. Option traders and analysts often use a combination of them.

6. Valuation, P/L and Greeks

This chapter explains the results of individual leg and total portfolio valuation, profit or loss and Greeks. It covers the areas which were marked as 3 and 4 in [chapter 2](#).



Per Share vs. Total

All the results covered in this chapter, except Price in column I, can be displayed either per share or as total dollar amounts for your particular position sizes. The selection is made in the combo in **cell L4**, which is labelled "Show Values & Greeks".

	H	I	J	K	L	M	N	O	P
1	macroption								
3	+1%	+5%	Div Yield	2.05%	Show Values & Greeks		-5%	-1%	Volatility
4	+0.1	+1	Int Rate	0.50%			-1	-0.1	+1% +5%
6	Initial CF	Price	Value	P/L		Gamma	Theta	Vega	Rho
8	-1,075.00	0.65	322.53	753.47	39.73	4.27	-10.33	62.71	-14.68
10	2,450.00	1.48	-739.19	1,710.81	90.58	-9.07	15.68	-111.67	33.48
12	2,325.00	2.14	-1,070.59	1,254.41	-140.95	-13.36	15.38	-143.08	-48.21
14	-975.00	0.62	307.66	-667.34	50.76	6.77	-8.63	75.24	17.50
16									
18	2,725.00		-1,179.58	1,545.42	-39.33	-11.40	12.10	-116.80	-11.91

Each column except Price also has the total portfolio sum displayed at the bottom in row 18. It is simply the sum of the five numbers above.

If you are working with a strategy which has different position sizes for different legs and have chosen to display the results per share, row 18 shows the sums as they would be if each leg's position size was one share, which might be a bit misleading sometimes. It is best to select "Total" in the combo in such cases.

Now we will cover the individual columns from left to right.

Initial CF (Column H)

This column displays the initial cash flow. When it has positive sign, it means you have received money when entering the position, typically when selling. Negative sign means an initial cash outflow, typically when buying.

When the results are being displayed per share, this column is simply the initial price from column G, only applying the proper sign. When the results are being displayed as total values, it is also multiplied by the position sizes and contract multipliers.

Price (Column I)

This is the price of the asset for each leg. It is the only column which is unaffected by the L4 combo selection – it is always displayed per share. For options the price is calculated using the option pricing model based on the leg inputs and the parameters in rows 3-4. For legs which are long or short in the underlying, it is equal to the underlying price entered in cell G4.

Value (Column J)

This column shows the value of individual legs and the total portfolio. When displayed per share, it is equal to the previous column, only with negative sign for short positions. When displayed as total value, it is also multiplied by position size and contract multiplier.

Profit or Loss (Column K)

This is probably the most important column, showing the profit or loss, simply adding up the initial cash flow (column H) and current value (column J).

Delta (Column L)

Delta measures sensitivity of an option's price to changes in the underlying price. Mathematically, delta is the first derivative of option's price with respect to underlying price.

The number in column L means by how many dollars the particular leg's or total portfolio's value (per share or total) would change if the underlying price increased by one dollar.

Generally, delta per share can range from 0 to +1 for call options and from -1 to 0 for put options. Far out of the money options have delta close to zero, while deep in the money options have delta near +1 (calls) or -1 (puts).

Delta is the only Greek which is also displayed for legs which are long or short the underlying. Delta of a long underlying position is exactly +1 (obviously, when the underlying price increases by one dollar, the value of a long underlying position also increases by one dollar per share). Delta of a short underlying position is exactly -1.

If you test it and change the underlying price by one dollar, you will realize that often the actual change in price does not match the value which was displayed as delta. It is usually very close, but not always exactly the same. This is because delta itself also changes with underlying price and the value you see as delta at any moment is only valid at the particular (infinitely small) point. As soon as the underlying price starts moving, even by one cent, the delta can change too. This phenomenon is called convexity and besides options, you can also find it on other financial instruments, such as bonds. The next Greek, gamma, is very important for measuring and managing convexity of options.

Gamma (Column M)

Gamma measures sensitivity of delta to changes in the underlying price – it is “delta of delta”. Mathematically, it is the second derivative of option’s price with respect to underlying price. The number you see in column M shows how much the option’s delta changes when the underlying price increases by one dollar.

Although somehow less popular and less widely understood than delta, gamma is a very important Greek. Sometimes your option portfolio has delta near zero, suggesting it won’t be very sensitive to moves in the underlying price, but if gamma is far from zero, the delta will quickly change, making a “non-directional” position very directional. This is often the case – and can be very dangerous – when trading short volatility strategies such as short straddles.

Gamma can only reach positive values for long option positions; negative for short. Positive gamma means that, if the underlying price continues to move in the same direction, your profits will accelerate and your losses will gradually slow down. Negative gamma means the opposite – accelerating losses and profits slowing down. However, to make things even more complicated, gamma itself can change when the underlying price (or some other parameter) changes.

For individual options, gamma is highest approximately (but not necessarily exactly) at the money and converges to zero on both tails (far out of the money and deep in the money).

Positions directly in the underlying asset have zero gamma (as their delta is constant at +1 for longs and -1 for shorts). The simulator does not display gamma for these.

Theta (Column N)

Theta measures sensitivity of the option’s price to passing time. Mathematically, theta is the first derivative of option’s price with respect to time to expiration. The number you see in column N shows how much the option’s price changes in one day (if the valuation date in cell C3 increases by one day).

For example, if an option's price is 1.50 today and theta is -0.03, the option's price will be 1.47 tomorrow (if the other parameters like underlying price and volatility remain constant, which rarely happens in real world).

In an environment of low interest rates and with low dividends, theta is negative for long option positions (options lose value with passing time).

Note that some trading platforms or software packages may show theta with opposite sign.

Vega (Column O)

Vega measures sensitivity of the option's price to changes in volatility. Mathematically, vega is the first derivative of option's price with respect to volatility. The number you see in column O shows how much the option's price changes when implied volatility rises by one percentage point per annum. The units of vega are not percentages – like the other Greeks, they are the same as option's and underlying price – usually dollars per share.

Vega is positive for long option positions (both calls and puts); negative for short. Other things being equal, vega is highest for at the money options and converges to zero on both tails (far out of the money and deep in the money). Vega also generally declines with passing time (options with longer time left to expiration have higher vega than similar options with less time to expiration).

Rho (Column P)

Rho measures sensitivity of the option's price to changes in the risk-free interest rate. Mathematically, rho is the first derivative of option's price with respect to interest rate.

The number you see in column P shows how much the option's price changes when the risk-free interest rate rises by one percentage point per annum.

Long calls and short puts have positive rho. Short calls and long puts have negative rho. Rho is generally greater (in absolute terms – more positive or more negative) for options with longer time to expiration.

In an environment of low and relatively stable interest rates, rho is usually less used and watched than the other Greeks, because the impact of interest rate changes on option prices is much smaller than impact of the other variables (such as volatility or underlying price). However, in times of double digit and volatile interest rates, rho can become very important.

Expired Options

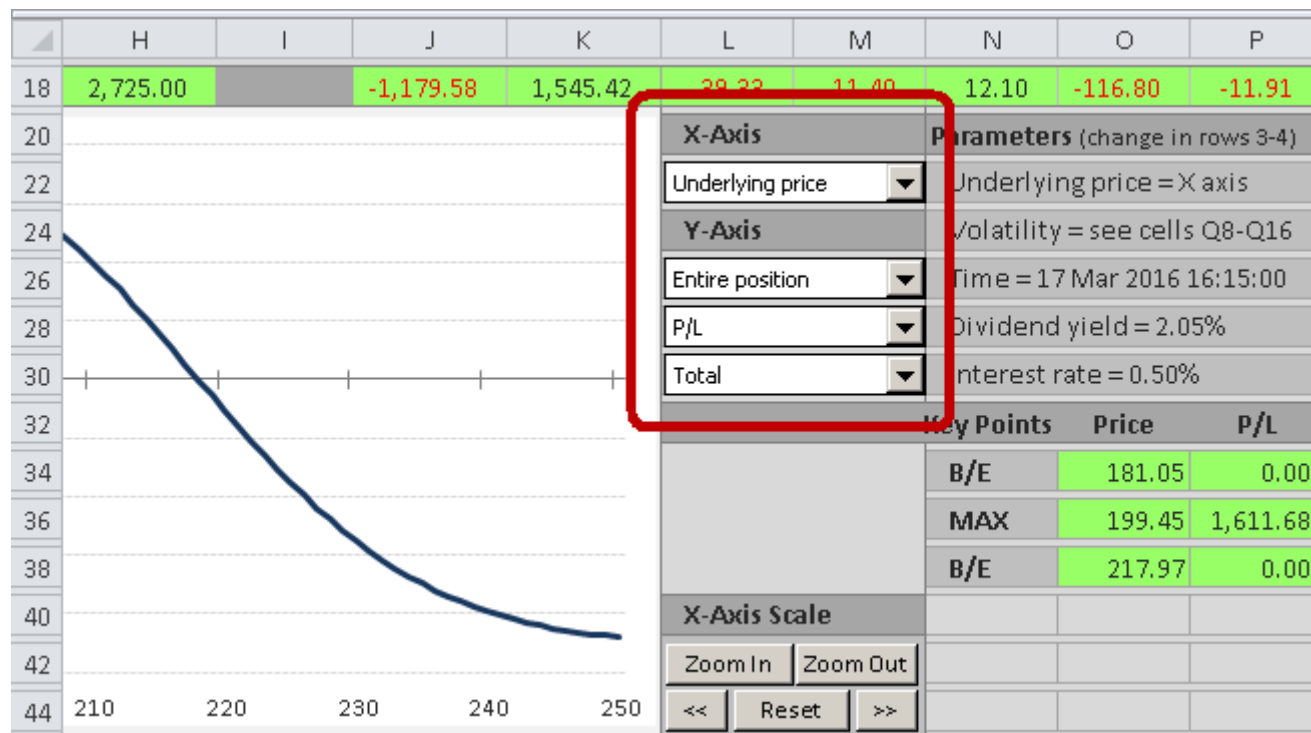
If the valuation date and time (cells C3-C4) are greater than or equal to an option's expiration date and time (the option has expired), The Greeks for the particular leg will not be shown. They are all zero.

7. Chart and Simulations

The results area covered in the [previous chapter](#) shows option prices, profit or loss and Greeks for a particular set of parameters, typically for the current date, underlying price and volatility. But often you also want to see how your position would behave under different scenarios and market conditions, what happens if the underlying price drops by a certain amount or if volatility increases, at what particular price or volatility level you make maximum profit or loss and how much, where and when you break even, or how the delta will change with passing time. This is where the simulator becomes much more powerful.

Setting X and Y Axes

The chart can show the effect of the individual parameters (underlying price, volatility, time, dividend yield and interest rate) on the position's value, profit or loss, or any of the Greeks. This can be shown either for the entire position or for an individual leg, either per share or total. The settings are controlled on the right side of the chart, in the area covering cells L20-M44.



The first combo (cell **L22-M22**, labelled “X-Axis”) selects which of the parameters will be the changing one. For example, if you want to study the effects of underlying price, select “Underlying price”. If you want to analyse how the profits and Greeks will change with passing time, select Time, and so on.

While the first combo selects “effect of what”, the three combos below, labelled “Y-Axis”, select “on what”.

In the **combo in cells L26-M26** select if you want to see the results for the entire position or an individual leg.

In **cells L28-M28** select if you want to plot the position’s value, profit or loss, or one of the Greeks.

Finally in **cells L30-M30** select if you want to see the results per share or total for your position size.

Changing Parameters

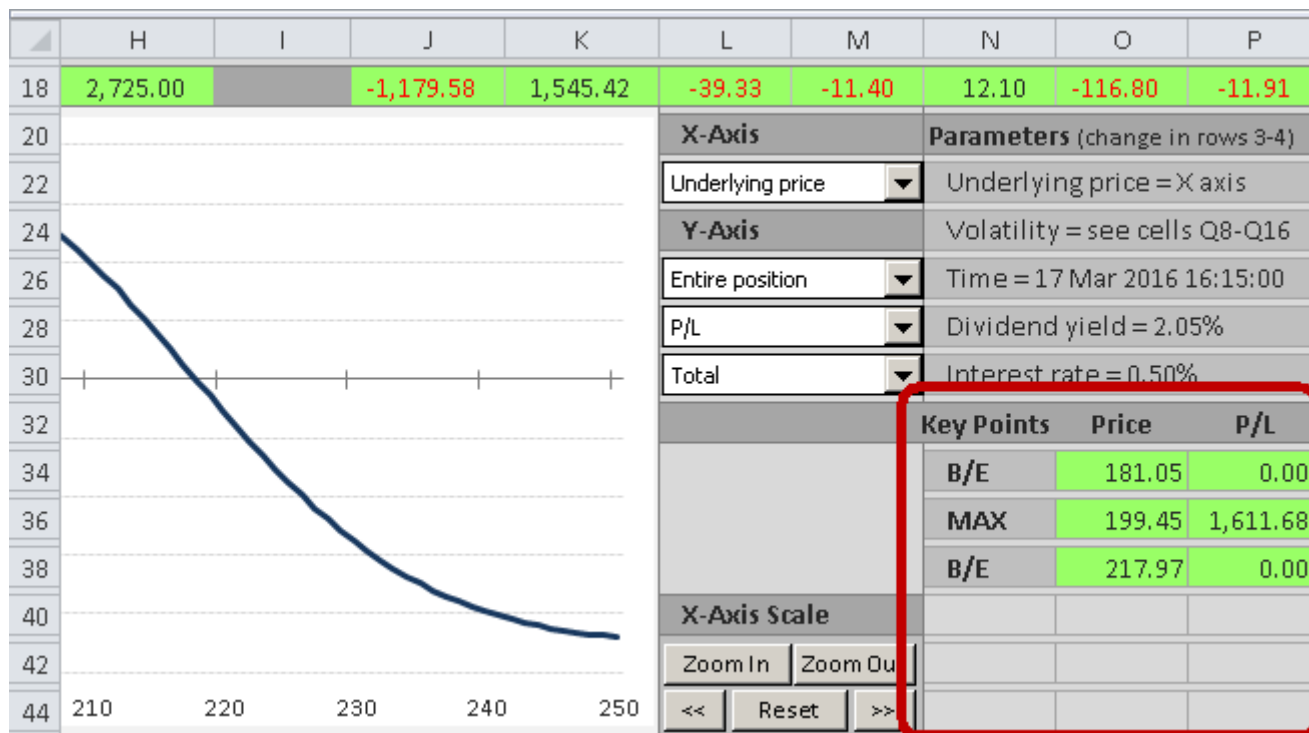
While one of the parameters is displayed on the X-axis and you can see its effect in the chart, all the other parameters remain constant and equal to what you have set in rows 3-4 (detailed instructions in [chapter 5](#)). For convenience you can also see their values in the summary in cells N20-P30.

Changing any of these parameters in rows 3-4 will update the chart, enabling you to study more complex relationships between the parameters and their effects. For example, when you are displaying the effect of underlying price on profit or loss and you change the valuation date by clicking on the “+1d” button in cell D3, you will see how the profit/loss profile changes with every passing day and how your break-even points, maximum profit and maximum loss shift.

	N	O	P
40	12.10	-116.80	-11.91
Parameters (change in rows 3-4)			
▼	Underlying price = X axis		
	Volatility = see cells Q8-Q16		
▼	Time = 17 Mar 2016 16:15:00		
▼	Dividend yield = 2.05%		
▼	Interest rate = 0.50%		
	Key Points	Price	P/I

Break-Even, Maximum and Minimum Points

Option traders often want to know the particular price or volatility level where their position breaks even or where it would reach maximum profit or maximum loss. These values, labelled “Key Points”, are displayed in cells N32-P44.



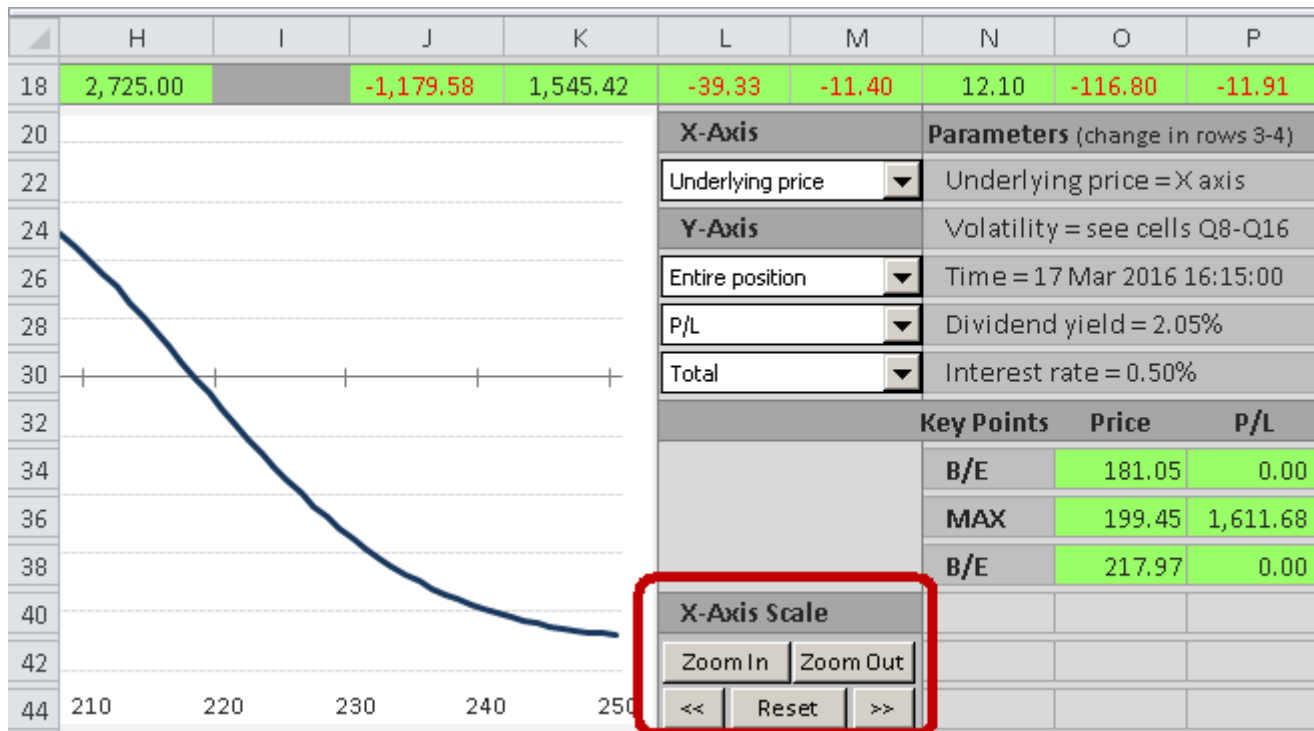
The first column (N) displays the type of point: “B/E” for break-even, “MAX” for maximum profit or smallest loss, “MIN” for maximum loss or smallest profit. The second column (O) displays the particular parameter’s value (e.g. underlying price) and the last column (P) shows the Y axis value (e.g. profit or loss).

The values you see are always based on the chart’s current settings. While most people are primarily interested in the effect of underlying price on their profit or loss, you can also display break-even, maximum and minimum points for delta or another Greek, based on underlying price or any other parameter. For example, when the chart is showing the effect of passing time on delta, the Key points area will show dates in column O (in place of underlying price) and delta in column P (in place of P/L).

Note that the Key points only cover the current X-axis range and if you have the chart zoomed in, some of the important points may be out of range and not shown. In such case just reset or zoom out the chart’s X-axis.

X-Axis Scale

You can control the chart's X-axis using the buttons in the bottom right corner of the chart, in cells L40-M44. Changing any of the legs or chart parameters resets the scale, while changing the option pricing parameters does not.



Volatility Input

In most cases when working with the simulator you can choose to work with different implied volatility levels for individual legs, allowing you to model skew. However, when displaying the effect of volatility in the chart ("Volatility" selected in the combo in L22-M22) the same volatility always applies to all legs, even if you have "Flat volatility" unchecked in cell Q3.

8. Option Pricing Model and Assumptions

The simulator uses the **Black-Scholes option pricing model**, which was first introduced by Fischer Black and Myron Scholes in [The Pricing of Options and Corporate Liabilities, 1973](#) and extended by Robert C. Merton in [Theory of Rational Option Pricing, 1973](#) to also work with dividends.

This model is the gold standard in finance, has greatly contributed to the expansion of option trading and earned its authors the Nobel Prize in 1997 (namely Merton and Scholes, as Black was unfortunately ineligible due to his death in 1995).

That said, like every theoretical model it has limitations and may deviate from the reality of trading in some ways. When using the model, **it is important to be aware of its assumptions and their implications for the model's accuracy and applicability.**

American vs. European Options

The calculations and charts in the simulator, as well as the original Black-Scholes-Merton model, apply to European options. In other words, it is assumed that an option can be exercised only at expiration and not before.

In reality, the difference between pricing of European and American options is negligible in most cases and therefore **it is accurate enough to price American options using the model.** Exceptions include some in the money call options on high dividend yield stocks or in the money puts in a high interest rate environment, especially when close to expiration.

Dividends

The original Black-Scholes option pricing model ([Black, Scholes, 1973](#)) assumes that the underlying security does not pay any dividends. In other words, dividends don't enter option price calculation in any way.

The simulator uses the expanded version of the model ([Merton, 1973](#)) which can price options on securities that pay a dividend. The calculation assumes that the underlying security pays a **continuous** dividend at the rate you set as entry parameter in cell J3.

Note: If you want the exact original Black-Scholes model, just set dividend yield to zero.

Convenience Yield and Forex Options

You can also use the simulator to work with options on assets that generate other kinds of cash payments or represent other benefits to their holders (while option holders don't get these payouts or benefits). This is known as **convenience yield**. Simply enter it as parameter in place of dividend yield in cell J3.

You can also use the simulator for **forex options**, where two interest rates (of the two currencies) enter the calculation, one of them as interest rate and the other in place of dividend or convenience yield. This is in line with the Garman-Kohlhagen forex option pricing model ([Garman, Kohlhagen, 1983](#)), which is mathematically identical to Merton's dividend extension of the Black-Scholes model.

Other Assumptions

The following are other assumptions made for both the original Black-Scholes-Merton model and the calculations in the simulator.

- **No riskless arbitrage** opportunity exists.
- There are **zero transaction costs**.
- There are **no restrictions to short selling** of the underlying security.
- You can **buy or sell any quantity** of the underlying security, even a fraction (e.g. 0.1843 shares).
- You can **borrow or lend cash**, in any quantity, at a **constant risk-free interest rate** (this is the interest rate parameter which you enter in cell J4).
- Future direction of the price of the underlying security can't be predicted. The price makes a "**random walk**". This also means that:
 - Returns are normally distributed.
 - Prices are log-normally distributed.

Weaknesses and Criticism of the Black-Scholes Model

Some of the assumptions above are clearly invalid or questionable in real world. You can find many papers and articles discussing the weaknesses of the Black-Scholes model. There are other option pricing methods which can be more suitable in many situations.

Nevertheless, the purpose of this guide is not to discuss whether the Black-Scholes model is a good way of pricing options. It assumes that you want to use it – for whatever reason – and explains how to do it using the simulator.

9. Formulas Used

Parameters

S_0 underlying price (USD per share)

X strike price (USD per share)

σ volatility (% p.a.)

r continuously compounded risk-free interest rate (% p.a.)

q continuously compounded dividend yield (% p.a.)

t time to expiration (% of year)

Calculation of Option Prices

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + t\left(r - q + \frac{\sigma^2}{2}\right)}{\sigma \sqrt{t}}$$

$$d_2 = d_1 - \sigma \sqrt{t}$$

$$C = S_0 e^{-qt} * N(d_1) - X e^{-rt} * N(d_2)$$

$$P = X e^{-rt} * N(-d_2) - S_0 e^{-qt} * N(-d_1)$$

... where $N(x)$ is the standard normal cumulative distribution function.

Note: In the original Black-Scholes model, which doesn't account for dividends, the equations are the same except:

- There is just S_0 in place of $S_0 e^{-qt}$
- There is no q in the formula for d_1

Therefore, if dividend yield is zero, then $e^{-qt} = 1$ and the models are identical.

Calculation of the Greeks

$$\text{Call delta} = e^{-qt} * N(d_1)$$

$$\text{Put delta} = e^{-qt} * (N(d_1) - 1)$$

$$\text{Gamma} = \frac{e^{-qt}}{S_0 \sigma \sqrt{t}} * \frac{1}{\sqrt{2\pi}} * e^{\frac{-d_1^2}{2}}$$

$$\text{Call theta} = \frac{1}{T} \left(- \left(\frac{S_0 \sigma e^{-qt}}{2\sqrt{t}} * \frac{1}{\sqrt{2\pi}} * e^{\frac{-d_1^2}{2}} \right) - r X e^{-rt} N(d_2) + q S_0 e^{-qt} N(d_1) \right)$$

$$\text{Put theta} = \frac{1}{T} \left(- \left(\frac{S_0 \sigma e^{-qt}}{2\sqrt{t}} * \frac{1}{\sqrt{2\pi}} * e^{\frac{-d_1^2}{2}} \right) + r X e^{-rt} N(-d_2) - q S_0 e^{-qt} N(-d_1) \right)$$

... where T is the number of days per year (calendar or trading days, depending on preferences)

$$\text{Vega} = \frac{1}{100} S_0 e^{-qt} \sqrt{t} * \frac{1}{\sqrt{2\pi}} * e^{\frac{-d_1^2}{2}}$$

$$\text{Call rho} = \frac{1}{100} X t e^{-rt} * N(d_2)$$

$$\text{Put rho} = -\frac{1}{100} X t e^{-rt} * N(-d_2)$$

Note: The term $\frac{1}{\sqrt{2\pi}} * e^{\frac{-d_1^2}{2}}$ that you can see in several formulas is the standard normal probability density function.

10. Calculation Area and Functions Used

The various calculations powering all the simulator's features are done in rows 46 and below in the sheet "OSS". You don't need to understand these to be able to use the simulator effectively. That said, this chapter provides basic orientation in the calculation area for those who wish to see how the pricing model is implemented in Excel and perhaps customize the spreadsheet. Do not change anything in this area unless you know what you are doing and want to change how the simulator works. It is strongly recommended to always keep a backup copy of the original file.

Structure of the Calculation Area

Option prices and Greeks are calculated for all the five individual legs, using the parameters and the Black-Scholes option pricing model explained in chapters [8](#) and [9](#).

Row 48 calculates these for the current parameters, with results being displayed in the green cells in rows 8-18. **Rows 50-150** calculate the same for different points on the chart's X-axis, with one parameter changing and the others constant, depending on the selection in the combo in cells L22-M22.

The **parameters** are listed in columns A-E, **leg 1** calculations in columns F-AD, and the **other legs** further to the right in columns AF-BD, BF-CD, CF-DD and DF-ED, respectively.

Columns EF-EH sum up all the legs and calculate the totals for the entire position, depending on the chart's settings.

Columns EI-EN contain calculations used to determine the break-even points, minimums and maximums.

Because the exact break-even or extreme point often falls in between two ticks of the X axis and the functions are often quite complex and non-linear, more accurate values are calculated in **rows 151 and below**. The formulas in most columns are the same in all rows; it is mostly the parameters which change.

Excel Functions Used

Most of the operations and Excel functions used are basic mathematics (addition, subtraction, multiplication, division, POWER, SQRT, EXP, LN).

The part that might be unfamiliar to many people is calculation of the **cumulative distribution functions** – the $N(d_1)$, $N(d_2)$, $N(-d_1)$, and $N(-d_2)$ from the formulas in the [previous chapter](#)).

It is done in columns O-R (and the corresponding columns for the other legs) using the **NORM.DIST** function. This function was named NORMDIST, without the dot, in older versions of Excel.

NORM.DIST for Cumulative Distribution Functions

There are 4 parameters to set with Excel NORM.DIST function:

- X – link to the cell with d_1 or d_2
- Mean – in our case 0, because it's standard normal distribution
- Standard deviation – in our case 1, for the same reason
- Cumulative – in our case TRUE, because it is cumulative

Example for $N(d_1)$ in cell O48: `NORM.DIST(M48,0,1,TRUE)`

... where M48 is the cell that calculates d_1 .

Note: There is also the NORM.S.DIST function in Excel, which is the same as NORM.DIST with mean = 0 and standard deviation = 1. You can use either.

NORM.DIST for Probability Density Functions

You can also use NORM.DIST to calculate the probability density functions which enter the calculation of gamma, theta, and vega. In the spreadsheet I calculated them manually in column W (and AW etc. for the other legs):

$$\frac{1}{\sqrt{2\pi}} * e^{\frac{-d_1^2}{2}}$$

In cell W48: $\text{EXP}(-1 * \text{POWER}(M48, 2) / 2) / \text{SQRT}(2 * \text{PI}())$

... where M48 is again the cell that calculates d_1 .

You would get the same result using the NORM.DIST function with the following settings:

- $X = -d_1$ (minus and link to the cell with d_1)
- Mean = 0
- Standard deviation = 1
- Cumulative = FALSE (this is the difference vs. the cumulative distribution functions above)

In Excel: $\text{NORM.DIST}(-M48, 0, 1, \text{FALSE})$

Auto X

Besides the main sheet there are also calculations in the sheet “Auto X”, which calculates a “nice” X axis scale (so it goes in round figures like 5, 10, 15 and not 4.78, 7.41 etc.). It also handles the X axis control buttons (recalculates the scale when you click “Zoom In”, “Zoom Out” etc.).

11. Common Technical Problems

The following remarks address the most common technical problems people encounter when using the simulator. If it's not working as expected or you see errors, you may find the cause and solution here. If not, please send an email to support@macroption.com.

You See Errors Such As #NAME

This most likely means that **you are trying to run the default version of the simulator (OSS.xlsm) in an older version of Excel.**

As was already discussed in chapters 1 and 10, the simulator uses the NORM.DIST function, which changed in Excel 2010. In previous versions there was a similar function named NORMDIST and unfortunately older versions of Excel do not recognize the new function. Therefore the default version of the simulator does not work in Excel 2007 and older.

If you have Excel 2007 or older, you need to use the other version, OSS_for_Excel_97-2007.xls.

File Does Nothing When You Try to Click on Things

This most likely means that **your Excel is blocking calculations or macros.** It is a measure designed to protect your computer from potentially harmful files and macros, especially those downloaded from the internet. The behaviour depends on your Excel version and on your particular security settings.

Check if there is a notice at the top, in place or below your Excel main menu. The wording differs, but it is often something like "This file originated from the internet and may be unsafe" or "Macros have been disabled" or similar. There is usually a button at the end of the notice, labelled "Enable content", "Enable calculations" or "Enable macros". **Clicking on this button should resolve the issue** and allow you to change cells and perform calculations.

12. References

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13. Contact and Terms of Use

Contact

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